Exact K2P and K3P Tree–Theta-Trinet Collisions

Technical summary

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Exact finding. A binary semi-directed strict level-two theta trinet with a genuine nontrivial 3-blob shares exact K2P and K3P distributions with a three-star tree. The K2P construction is a counterexample to Lemma 5.6 and the K2P part of Corollary 5.8 in arXiv:2607.12919v2. Version 3 removes those K2P claims, identifies the leaf-order obstruction, and leaves high-level K2P and K3P distinguishability open; the two exact collisions answer both questions negatively. They do not affect the corrected paper's JC or level-one results.

Common topology. Use

$$\rho \to 1, \rho \to u,\ u \to p, q,\ p, q \to r_2,\ p, q \to r_3,\ r_2 \to 2,\ r_3 \to 3,$$

with both inheritance probabilities 1/2. Root suppression gives paths $p - u - q$, $p - r_2 - q$, $p - r_3 - q$ and three incident leaf edges. For the explicit symmetric witnesses, U labels $u \to p$, V labels $u \to q$, S labels both $p \to r_2, p \to r_3$, and T labels both $q \to r_2, q \to r_3$; the other four arcs carry K. The rank and implicit-function calculations instead use independent parameters on all nine effective semi-directed edges and both inheritance probabilities.

Simple exact K2P witness. In order (A, C, G, T), assign

$$K = (1, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}), \quad U = (1, \tfrac{4}{5}, \tfrac{19}{30}, \tfrac{4}{5}), \quad V = (1, \tfrac{7}{240}, \tfrac{239}{360}, \tfrac{7}{240}),$$
$$S = (1, \tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{4}), \quad T = (1, \tfrac{1}{3}, \tfrac{1}{27}, \tfrac{1}{3}).$$

Use K on the four common K-edges $\rho \to 1, \rho \to u, r_2 \to 2, r_3 \to 3$, with the explicit U, V, S, T placement above. Every transition entry is positive; the smallest is 1/120. For $x = y + z$, these common edges contribute $K_x K_{y+z} K_y K_z = K_x^2 K_y K_z$. The four retained-parent choices $(p, p), (p, q), (q, p), (q, q)$ give, respectively, the four terms in

$$M_{y,z} = \tfrac{1}{4}(S_y S_z U_x + S_y T_z U_y V_z + T_y S_z U_z V_y + T_y T_z V_x).$$

With $\eta = \sqrt{71}$,

$$P = (1, \tfrac{151}{36\eta}, \tfrac{107}{162}, \tfrac{151}{36\eta}), \quad R = (1, \tfrac{\eta}{40}, \tfrac{31}{120}, \tfrac{\eta}{40}),$$

and exact arithmetic gives $M_{y,z} = P_{y+z} R_y R_z$. The tree vectors are

$$\alpha = (1, \tfrac{151\eta}{10224}, \tfrac{107}{648}, \tfrac{151\eta}{10224}), \quad \beta = \gamma = (1, \tfrac{\eta}{80}, \tfrac{31}{240}, \tfrac{\eta}{80}).$$

All 64 Fourier and pattern probabilities agree exactly; the minimum is $1188799/79626240 = 0.01492973924\ldots$. The source invariant Q is zero. A graph-based verifier literally deletes the two unselected reticulation arcs, reconstructs the four monomials, and independently confirms all 64 probabilities by ordinary-state Markov pruning.

Proof diagnosis. The level-one sign factorization is valid after placing the active reticulation leaf in a designated position. In the level-two induction a child can retain a reticulation adjacent to another leaf. Positivity after a child-specific relabelling does not imply positivity of the fixed-order child invariant in the parent expansion. Version 3 now records this obstruction. On the exact edgewise continuous-time witness, one child has $Q = -1.919971072382827\ldots \times 10^{-9}$ in the parent order and $Q = 3.428488326525925\ldots \times 10^{-9}$ after the favorable swap. A separate fully specified rational theta point has $Q < 0$ in all six orders.

K2P geometry and edgewise continuous time. Here "edgewise" means that every edge separately has a positive-rate symmetric generator; no common generator, rate ratio, molecular clock, or global temporal compatibility is imposed. The affine K2P space has dimension 9 and the tree model dimension 6. At the simple collision, the fixed theta map has determinant

$$-\frac{7^2 11^2 19\,107\,151^2\,15013}{2^{60}3^{25}5^{10}} \neq 0.$$

It also has rank 9 at the edgewise strictly continuous-time collision. Hence the local semi-directed theta collision locus has dimension 17 and codimension 3: generic theta parameters are distinguishable, but full distinguishability fails. An exact symmetric ansatz contains a smooth local six-dimensional semialgebraic family of positive collisions. The edgewise continuous-time witness satisfies $g > s^2$ on every edge, has minimum eigenvalue 1/16 and minimum rate margin 11/900, and is verified by direct Markov pruning.

K3P companion. Let $h = 5^{-1/4}$. Use the same fixed edge placement and the same four-switching matrix M, now with

$$U = (1, h/3, h, 1/3),\ V = (1, h, h/3, 1/3),\ S = (1, 3h^2/4, 1/4, 3/10),\ T = (1, 1/4, 3h^2/4, 3/10).$$

This mixture factors with

$$B = (1, h^2/2, h^2/2, h^2/2), \quad P = (1, (5h^3 + h)/4, (5h^3 + h)/4, h^2).$$

The fixed theta map has rank 15 with

$$\det J_* = \frac{h(10h^2 + 1)}{2^{61}3^4 5^{14}} > 0.$$

Its local collision locus has dimension 23 and codimension 6. A real-analytic implicit-function argument, with Jacobian and tangent data verified exactly, moves the collision into the edgewise cone where all three K3P rate classes are positive. The theta admits no tree-child rooting, so these collisions do not contradict results restricted to tree-child or strongly tree-child level-two networks.

Status and replay. The canonical manuscript, exact certificates, and verifier package are available in the versioned GitHub snapshot (v1.1.0). With Python 3.10 or newer, run `python3 verify_k2p_displayed_trees.py` for the graph reconstruction and direct-pruning audit, or `python3 verify.py` for every check; the latter ends with `ALL EXACT CHECKS PASSED`.